



Efficiently drawing linear graphs

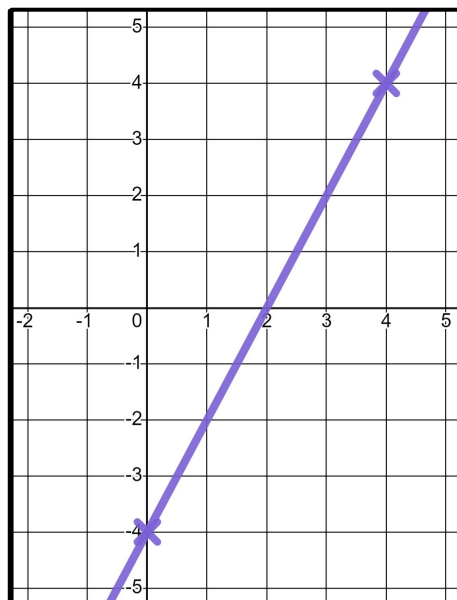
Task A: Plotting linear graphs

1) In each case fill in the missing coordinate and draw the respective linear graph.

a) $y = 2x - 4$

$(0, -4)$

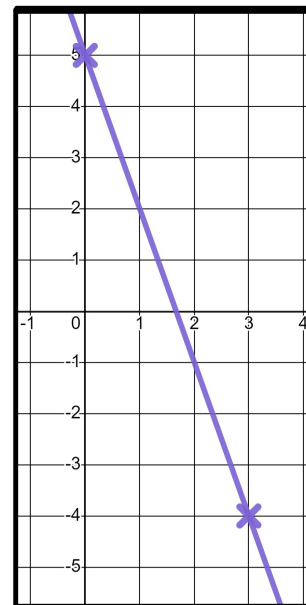
$(4, 4)$



b) $y = 5 - 3x$

$(0, 5)$

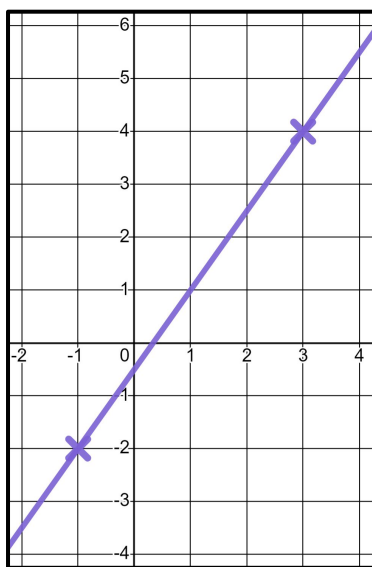
$(3, -4)$



c) $y = \frac{3x - 1}{2}$

$(-1, -2)$

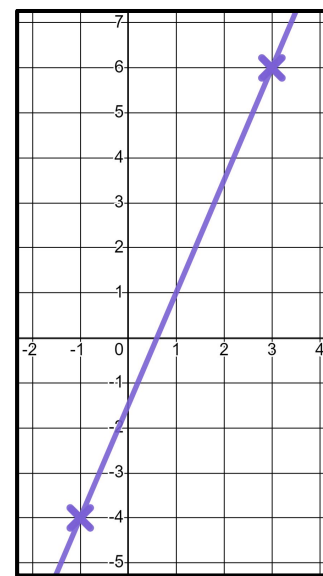
$(3, 4)$



d) $y = \frac{5}{2}x - \frac{3}{2}$

$(-1, -4)$

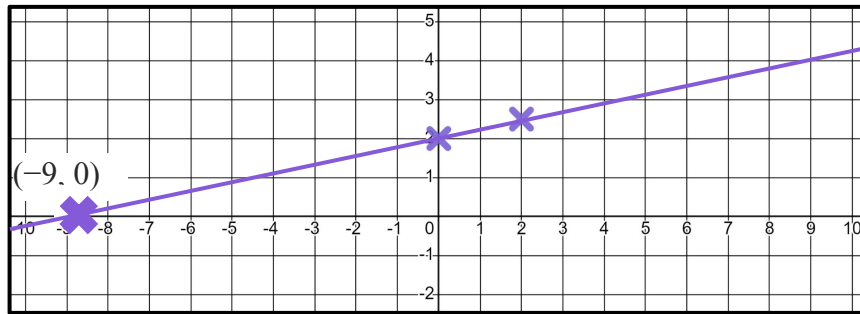
$(3, 6)$



2) Alex plots the linear graph of $y = 2 + \frac{x}{4}$ using these two coordinates.

$(0, 2)$

$(2, 2\frac{1}{2})$



Use the coordinate $(-9, 0)$ to demonstrate that they are wrong and write a sentence advising what they might do better.

$$2 + \frac{-9}{4} \neq 0 \text{ so } (-9, 0) \text{ can not be a coordinate on the line } y = 2 + \frac{x}{4}$$

You might have written:

"Start with two more distant coordinates to increase accuracy"

Task B: Efficiency in plotting

1) Find the coordinates for when $x = 0$ and when $y = 0$ thus draw the linear equations.

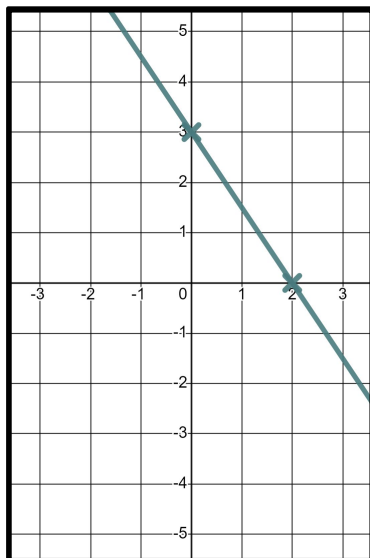
a) $3x + 2y = 6$

When $x = 0$

(0, 3)

When $y = 0$

(2, 0)



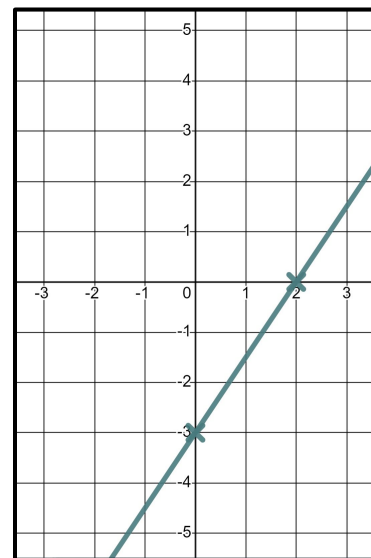
b) $3x - 2y = 6$

When $x = 0$

(0, -3)

When $y = 0$

(2, 0)



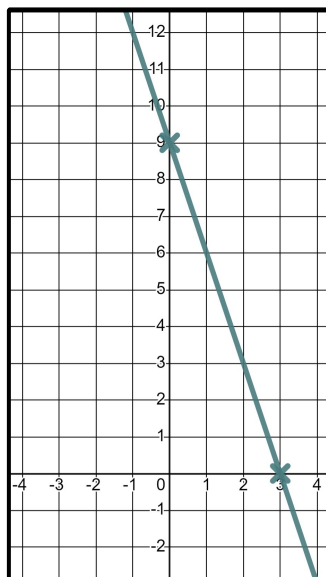
c) $9x + 3y = 27$

When $x = 0$

(0, 9)

When $y = 0$

(3, 0)



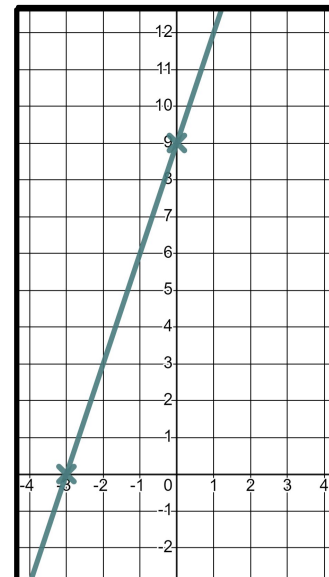
d) $3y - 9x = 27$

When $x = 0$

(0, 9)

When $y = 0$

(-3, 0)



2) For each graph find two x -values that will give you integer y -values and draw the linear equations.

a) $y = 0.8x - 3$

(0, -3)

(-5, -7)

(5, 1)

Then because 0.8, 1.6, 2.4, 3.2, 4, ...you know you'll hit another integer for every change of 5 in x

b) $y = 3.25 - 1.25x$

(1, 2)

(5, -3)

(-3, 7)

Then because 1.25, 2.5, 3.75, 5, ...you know you'll hit another integer for every change of 4 in x

